

ANIMASAUN DAVID

Senior Site Reliability Engineer | SRE · DevSecOps · Cloud Platform Reliability

daviddmilehin739@gmail.com | +234 802 934 1783 | +234 903 195 3494 | Lagos, Nigeria | AWS Certified Solutions Architect

PROFESSIONAL SUMMARY

Site Reliability Engineer with production ownership of security-critical data platforms on AWS. I keep high-volume pipelines correct and observable: diagnosing silent data-loss failures, removing memory and API-concurrency bottlenecks, and replacing invisible errors with structured signal engineers can act on.

DevSecOps depth across agent identity and gateway authorization (Bedrock AgentCore with Microsoft Entra ID, OAuth2 machine-to-machine, MCP), vulnerability-management data pipelines, Infrastructure as Code (Terraform, CloudFormation, CDK), CI/CD on Jenkins, containerized workloads, cloud security hardening, cost guardrails, and incident response. AWS Certified Solutions Architect.

TECHNICAL SKILLS

Reliability & Operations: Incident response and on-call, root-cause and failure-mode analysis, retry/backoff and idempotency design, rate limiting and quota enforcement, memory and capacity profiling, fault-tolerant multi-AZ design, disaster recovery, runbooks and handoffs

Observability: CloudWatch metrics, logs and alarms, AWS X-Ray tracing, structured logging and diagnostic instrumentation, per-job failure ledgers with step attribution, Jenkins build telemetry, Tableau operational dashboards

CI/CD & IaC: Jenkins multibranch pipelines, AWS CodePipeline, CodeBuild, CodeDeploy, Terraform, CloudFormation, AWS CDK, Vercel, Git, automated test gates, pinned reproducible runtimes

Containers & Compute: Docker, Amazon ECS in production, AWS Lambda, EC2, Auto Scaling, ELB/ALB, Step Functions, EventBridge. Kubernetes and Amazon EKS: working knowledge, not yet operated at production scale

AWS & Networking: S3, DynamoDB, RDS and Aurora, ECR, Cognito, SES, Textract, Bedrock, SageMaker, API Gateway, CloudFront, Route 53, Cost Explorer, Budgets; VPC design, subnets, route tables, security groups, NACLs, VPN, Direct Connect. GCP working familiarity

Security & Identity: Tenable.io and pyTenable, vulnerability-management pipelines, least-privilege IAM, KMS, SSE-S3/AES256, SSM Parameter Store, WAF, Shield, GuardDuty, CloudTrail, AWS Config, Microsoft Entra ID, OAuth2 client credentials and OIDC, JWT/JWKS and scope-claim validation, WebAuthn, CSP and security headers, CycloneDX SBOM

AI & Agent Platforms: Amazon Bedrock, Bedrock AgentCore and AgentCore Gateway, Model Context Protocol (MCP) servers and targets, RAG pipelines, AWS Textract, SageMaker

AI-Assisted Engineering: Claude Code for agentic development, debugging and refactoring; prompt engineering and prompt-template design; LLM-assisted incident analysis and root-cause investigation; agent tool design and evaluation

Languages: Python, Bash/Shell, TypeScript, JavaScript, SQL, Linux/Unix, boto3, AWS SDK v3

PROFESSIONAL EXPERIENCE

CORNERSTONE ONDEMAND | Site Reliability / DevSecOps Engineer

Jan 2025 to Present

Reliability and delivery ownership across the security engineering platform of a global talent-management SaaS provider: vulnerability-management data pipelines, an AI risk-assessment agent runtime, and the AWS and Jenkins infrastructure both run on.

Vulnerability Data Platform (Tenable.io to S3 to Tableau)

- **Own the production vulnerability export pipeline** serving roughly 19 product lines and 2.2M+ findings per run, feeding the risk dashboards the security organization uses to prioritize remediation.
- **Root-caused a silent data-loss regression** in which exports completed cleanly with zero rows and no error. Ran a controlled comparison against the legacy pipeline on identical same-day data (1,174 rows versus 0), eliminated severity filters, state, time window and API concurrency as causes, and isolated the fault to a Unicode en dash (U+2013) standing in for an ASCII hyphen between locally normalized asset tags and the raw `tag.<category>` filter key sent over the wire, confirmed by diffing both pipelines' HTTP export payloads.
- **Re-architected the export strategy** from per-product tag-filtered exports to a single asset-tag partition map plus two global state-scoped exports, cutting external export jobs per run from roughly 133 to 3, removing a fragile server-side filter from the critical path, and clearing the concurrency pressure behind the vendor's 3 to 5 concurrent-job ceiling.
- **Removed a multi-gigabyte OOM failure mode** by converting fully materialized in-memory result sets and string concatenation to generator streaming into temporary files, with boto3 multipart upload (64 MB threshold, 16 MB chunks, concurrency 5) and SSE-AES256 enforced on write.
- **Restored data-quality guarantees** lost in a rewrite: cross-file deduplication on composite finding keys, and removal of an over-restrictive severity-modification filter that had been zeroing out entire product lines and skewing downstream Tableau counts.

Agent Identity & Gateway Security (Bedrock AgentCore, Microsoft Entra ID)

- **Integrated Microsoft Entra ID as the OAuth2 identity provider for an Amazon Bedrock AgentCore Gateway**, standing up the machine-to-machine client-credentials flow so MCP clients including Claude Code authenticate with short-lived JWTs (3600 second expiry) instead of long-lived static credentials.

- **Debugged the full inbound-auth chain** across Entra app registration, application scopes and short names, required group membership, JWT audience and issuer alignment, and custom scope-claim validation on the gateway authorizer. Separated three causes of 403 rejection that had been producing identical symptoms: wrong scope claim, wrong audience, and missing group assignment.
- **Registered MCP servers as HTTP gateway targets** and wired the Entra credential provider into the agent identity layer, giving agent runtimes authenticated, scope-restricted tool access rather than open network reach.
- **Closed a least-privilege IAM gap** where token retrieval failed because no identity-based policy permitted `ssm:GetParameter`, keeping client secrets in SSM Parameter Store rather than in application code or environment files.

AI Risk-Assessment Agent Platform

- **Build and operate an AgentCore-based security risk-assessment agent runtime** in Python on AWS Bedrock, together with the data-sync services that keep its underlying risk corpus current.
- **Deliver the agent runtime through Jenkins multibranch pipelines** with reproducible pinned Python environments, promoting pre-flight dependency and credential checks into their own build stage so environment drift fails fast rather than mid-run.

Reliability Engineering, Observability & Incident Response

- **Corrected a misdiagnosed failure mode** that had shaped the system's design: a warning wrongly attributing zero-row exports to API throttling had justified a 30 second inter-product sleep and a three-attempt retry ladder burning roughly 2.5 minutes of dead wait per empty product. Re-gated retries on evidence of throttling rather than on empty results.
- **Replaced invisible failures with actionable signal**: per-product failure ledger with step attribution and tracebacks, filter-rate and asset-match diagnostics, distinct process exit codes for full, partial and total failure, and pre-flight Tenable and S3 health checks promoted to a Jenkins stage.
- **Provision infrastructure as code** with Terraform and CloudFormation, standardizing environments and enforcing least-privilege IAM, encryption at rest, and repeatable multi-account deployments.
- **Participate in on-call and incident response** for security platform services: triage from Jenkins and CloudWatch evidence, hypothesis-driven root-cause analysis, and written remediation handoffs other engineers execute against.

NIMBUS SOLUTIONS PROVIDER | *Lead Engineer, Cloud Infrastructure & SRE*

Aug 2024 to Present

Lead reliability, security and delivery-pipeline architecture for enterprise and startup client platforms on AWS.

- **Architect multi-stage delivery pipelines** on CodePipeline, CodeBuild, CodeDeploy and Jenkins with explicit build, test, security-scan and deploy gates, environment promotion, and defined rollback paths, so releases are repeatable rather than heroic.
- **Design security architecture for client AWS estates**: least-privilege IAM boundaries and access-control models, KMS and SSE-S3 encryption at rest, WAF and Shield at the edge, GuardDuty and CloudTrail for detection, AWS Config for drift, and secrets held outside application code.
- **Architect event-driven data and processing pipelines** using Lambda, EventBridge, Step Functions, S3 events and DynamoDB streams, orchestrating with Step Functions so retries, error branches and idempotency are explicit in the topology rather than buried in application code.
- **Codify infrastructure in CloudFormation and CDK**, standardizing multi-AZ VPC topology, subnet and security-group boundaries, auto-scaling policies, and parity across development, staging and production.
- **Build observability into delivery** with CloudWatch metrics, logs and alarms, X-Ray tracing across service hops, and structured application logging, reducing time-to-diagnosis on client incidents.
- **Advise on reliability and cost trade-offs** against the AWS Well-Architected Framework, covering service selection, high-availability targets, disaster-recovery posture and right-sizing.

SELECTED PROJECT

MAILVAULT / LTTR (getlttr.com) | *Cloud Architect & Lead Engineer, Web and Mobile*

2026

Serverless mail-digitization platform on AWS: users capture physical mail, Textract extracts the text, Bedrock summarizes and categorizes it, and caregivers can be granted delegated access. Next.js 16, React 19 and TypeScript on Vercel, fully AWS-backed, shipped as a PWA with a native iOS and Android client.

- **Architected the serverless stack and codified it in CloudFormation** (410 line template): Cognito user pool with Google OAuth federation and hosted UI, S3 document bucket, DynamoDB single-table design, and a scoped IAM policy, deployable as one parameterized stack per environment.
- **Built the document-processing pipeline**: presigned S3 upload, JPEG normalization ahead of OCR, AWS Textract extraction, then Bedrock inference for summarization, categorization and per-document chat, with results and conversation history persisted to DynamoDB.
- **Hardened the platform in depth**: strict Content-Security-Policy, HSTS with preload, frame-ancestors none, X-Frame-Options DENY, nosniff, Permissions-Policy, production HTTPS downgrade redirects, and JWT verification against Cognito's remote JWKS in edge middleware on every protected route.
- **Implemented abuse and cost controls**: DynamoDB sliding-window rate limiting (10 login attempts per IP per 15 minutes) and atomic per-user daily AI quotas, both built on conditional atomic counter updates so limits hold under concurrent requests.
- **Designed AWS cost guardrails**: EventBridge-scheduled Lambda over Cost Explorer with tiered daily thresholds (\$500 warning, \$1,000 danger, \$2,000 critical), DynamoDB alert deduplication so a crossed threshold pages once rather than hourly, SNS fan-out to Slack and

email, LLM-generated cause analysis, and Step Functions remediation gated behind resource tags and human approval with dry run as the default mode. AWS Budgets and Cost Anomaly Detection wired in as independent backup signals.

- **Shipped passwordless and biometric authentication** with WebAuthn, plus session guards, delegated caregiver access with invite, accept and revoke flows, and role-based access control for admin-only publishing.
- **Published a CycloneDX 1.5 Software Bill of Materials** covering every runtime and build dependency and each outbound network service, for supply-chain review.
- **Designed an internal AI gateway boundary** so Bedrock, Anthropic, OpenAI or self-hosted models sit behind a single provider interface, keeping model choice swappable without touching product flows, with API Gateway or ALB fronting public APIs, S3-event-triggered Lambda orchestration, and EventBridge for scheduled reminders. Specified ingestion as an asynchronous pipeline with visible processing states so users never block on a synchronous OCR request.
- **Authored the architecture trade-off analysis** comparing a managed AWS path against an open-source hybrid across cost, performance and operational complexity, and recommended a hybrid document classifier (layout signals, OCR text, rules, then LLM verification) with confidence thresholds and human-review fallback.
- **Designed the native mobile client** on React Native and Expo (Expo Router, NativeWind, expo-secure-store, native Face ID and Touch ID, EAS cloud builds and over-the-air updates) against the same unchanged API surface, so no backend or data migration was required.

EARLIER EXPERIENCE

TURING | AI Engineer (Contract)

Jun 2024 to Sep 2024

- Conducted Reinforcement Learning with Human Feedback on large language models to improve performance across complex codebases.
- Built testing frameworks to monitor and document model performance, surfacing regressions and validating improvements; curated fine-tuning datasets aligned to real-world scenarios.

CDLP HUB LTD | Software Engineer

Nov 2022 to Nov 2023

- Architected responsive web applications with React and Node.js on a component-based architecture, improving code reusability by 40%. Promoted from intern to full-time engineer.
- Optimized performance via code splitting, lazy loading and state management for 50% faster page loads; led flagship platform features that raised user engagement 35%.
- Implemented automated Jest testing pipelines at 85% coverage, cutting production defects by 60%; mentored junior developers and set frontend practices adopted as company standard.

COMMUNITY, CERTIFICATIONS & EDUCATION

Google Developer Groups on Campus: AI/ML Team Lead, Dec 2024 to Present. Mentor developers in cloud computing, DevOps, AI/ML and scalable infrastructure; deliver talks and hands-on labs on serverless, CI/CD, containerization and cloud security.

Other Faces of Tech: Frontend Engineer, Jul 2023 to Jan 2025. Built adaptive, high-performance web experiences with strong cross-browser compatibility.

Certifications: AWS Certified Solutions Architect Associate. AWS Certified Cloud Practitioner. AWS Partner Technical Accredited.

Education: B.Sc. Physics, University of Ibadan.